

A perturbation of the seven-eighths zero-free half-plane

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Abstract

We examine the numerical endpoint in the compensated-probe proof of the seven-eighths zero-free half-plane in [1]. The reflected-row estimate has room to increase the total prime length, while the high-side endpoint certificate has a fixed positive margin. We prove the detector and continuation argument at boundary $174999/200000 = 0.874995$, conditional on a fourth-moment estimate for the perturbed source. This note records the parameter changes, the needed input, and the estimates used at the boundary.

1 Statement and input

Let $F = \mathbb{Q}(\sqrt{-3})$. As in [1, Equation (1.1b)], let β_* be the supremum of $1/2$ and the real parts of zeros of primitive finite-order Hecke L -functions over F in the critical strip. We use the seven-eighths theorem of [1], so $\beta_* \leq 7/8$. We also use the arithmetic identities, detector estimates, reflected-energy lemma, additive Gram bound, and continuation criterion proved there. Section 2 checks the changes needed when its Part II geometry is perturbed. Section 3 checks the high comparison and the analytic domains.

The perturbed fourth-moment input asserts the existence of detector data whose source moments hold for every resulting bin with label $a \leq 7/8$, zero moment excess $\Delta = 0$, and total prime-slot length $1/6 + 3/100000$. It includes the same count parameters and polynomial height allowance as the original chosen-moment input. The slot data can be chosen fine enough for any positive moment mesh. This input is not established here.

Theorem 1. *Assume the perturbed fourth-moment input above. Every finite-order Hecke L -function over F and every Dirichlet L -function, including the Riemann zeta function, has no zero in*

$$\operatorname{Re} s > B := \frac{174999}{200000} = \frac{7}{8} - \frac{1}{200000}.$$

The usual pole at $s = 1$ for a principal character is allowed.

Put

$$\begin{aligned} t &= \frac{3}{100000}, & \ell &= \frac{1}{6} + t, & b &= \frac{1}{8}, \\ l_x &= \frac{17}{48} - \frac{t}{2}, & l_y &= \frac{23}{48} - \frac{t}{2}, & h &= \frac{13}{16} + \frac{3t}{2}, \\ M &= l_x + l_y = \frac{5}{6} - t. \end{aligned} \tag{1}$$

These lengths satisfy $l_y - l_x = b$, $M + \ell = 1$, and $h = 1 - l_x + \ell$. The general Mellin-exponent identity in [1, Part II, compensated low estimate] therefore remains

$$C(s) = \frac{l_x}{2} + s - 1 + \frac{h}{6} = s - \frac{11}{16}. \tag{2}$$

Choose a fixed even number K of disjoint prime slots, each of length ℓ/K , as in [1, Section 5, “Order of choices”]. The slot count may be enlarged to meet the same mesh requirements. The finite physical probe and its exact Poisson identity in [1] make sense with the lengths in (1). All its arithmetic factors are unchanged.

2 The low comparison

Lemma 2. *The normalized compensated probe of [1], formed with (1), satisfies for every $\varepsilon > 0$*

$$|J_\eta(Z)| \ll_{\eta,\varepsilon} Z^{3/16-t/4+\varepsilon}.$$

Proof. Fix a rescaled subset of slots of total logarithmic length $0 \leq d \leq \ell$. In the notation of the compensated low estimate of [1], its row and surviving-slot lengths are

$$M' = M - 2d, \quad \ell' = \ell - d.$$

The proof of the reflected-row norm in [1, Part II, compensated low estimate] first bounds one branch by M' and the other by $M' + (1 + 3\ell' - 2M')/4$, up to arbitrary small losses. Its general reflected-energy lemma permits these fixed bounded lengths. Substitution gives the squared row norm

$$\sum_{0 < N m \ll Z^{M'}} |B_{m,\sigma}^J(Z)|^2 \ll Z^{M' + ((d-1/6+5t)/4)_+ + \varepsilon}. \quad (3)$$

The additive Gram estimate is available throughout this range. Its scale conditions follow from

$$\begin{aligned} l_x - \ell &= \frac{3}{16} - \frac{3t}{2} > 0, & l_y - \ell &= \frac{5}{16} - \frac{3t}{2} > 0, \\ M - 2\ell &= \frac{1}{2} - 3t > 0, & l_y - \ell - \frac{11b}{6} &= \frac{1}{12} - \frac{3t}{2} > 0. \end{aligned}$$

Thus its P_a^2/Y' term is absorbed by $P_a^{1/6}$, exactly as in [1, Part II, additive Gram bound]. Cauchy–Schwarz, the count of rescaled tuples, and their coefficients add to the basic low exponent

$$-d + \left(\frac{d - 1/6 + 5t}{8} \right)_+.$$

This is nonpositive for $0 \leq d \leq \ell$. Indeed, its positive-part argument is negative until $d > 1/6 - 5t$, and thereafter $d - 1/6 + 5t < 8d$. The empty subset has exponent zero. The direct bound is consequently $Z^{l_x/2+b/12+\varepsilon}$.

The prime normalizer in [1, Part II, principal normalizer] is nonzero and has inverse $Z^{o(1)}$ for every fixed positive slot length. Finally,

$$\frac{l_x}{2} + \frac{b}{12} = \frac{3}{16} - \frac{t}{4}.$$

This proves the normalized estimate. □

The signal at the proposed boundary is

$$C(B) = \frac{3}{16} - \frac{1}{200000}, \quad C(B) - \left(\frac{3}{16} - \frac{t}{4} \right) = \frac{1}{400000} > 0. \quad (4)$$

3 The high comparison

We first record the extension of the Euler regions used by the contour arguments. The local factors below are those of [1, Equations (5.13a)–(5.13c)].

Lemma 3. *The full compensated correction is holomorphic, with the bounds needed for principal-row contour moves, on the second Euler region of [1] with its condition $\operatorname{Re} s \geq 7/8$ replaced by $\operatorname{Re} s \geq B$. The principal local quotient there satisfies*

$$G_p/H_p = -1 + O((Np)^{-B}).$$

The small-row contour can use the first Euler region with $\epsilon_0 = 37/100$.

Proof. The exact local defect formula in [1, local Euler lemma] bounds $H_p - 1$ in the original second region by $O(Q^{-363/200})$ for $p \nmid u$ and $O(Q^{-33/40})$ for $p \mid u$, where $Q = Np$. Every local exponent involving $\operatorname{Re} s$ has coefficient of absolute value at most six. Moving its lower boundary left by $1/200000$ therefore gives the respective bounds

$$O(Q^{-363/200+3/100000}) \quad \text{and} \quad O(Q^{-33/40+3/100000}).$$

The first exponent is less than -1 . The unramified product converges normally, while the ramified product has the same $O_\varepsilon((Nu)^\varepsilon)$ bound. For $u = 1$ its tail still tends to zero faster than $P_0^{-4/5}$. Enlarging the fixed excluded set therefore gives the same uniform nonvanishing of the principal product.

The four errors for the principal selected factor have exponents $-\operatorname{Re} s$, $-6 \operatorname{Re} z$, $4 - 5 \operatorname{Re} s - 6 \operatorname{Re} z$, and $1 - \operatorname{Re} w - 6 \operatorname{Re} z$. On the extended region they are all at most $-B$. The formula for G_p has no division by P_p or H_p , so the full finite selected-tuple correction remains holomorphic at their zeros. The bounds above and finite slot count give the all-height majorant used in the principal contour proof of [1].

For small rows, $\beta_* > B$ implies $\beta_* + e + 1/2 > 1 + 37/100$. The first Euler region of [1] allows any positive ϵ_0 , so its unchanged local estimates apply to the line $(\beta_* + e, 1/2, 17/50)$. The central bin contours already lie in its region with $\epsilon_0 = 10e$. These are the only analytic domain uses of the original $7/8$ lower bound in the shared contour lemmas. Repeating their proofs with the extended region gives the same principal, bin, and outer-row contour estimates. $\square$

Lemma 4. *Under the perturbed fourth-moment input, if $B < \beta_* \leq 7/8$, then for some $\sigma > 0$ independent of the target character,*

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma},$$

where f_η is the principal Mellin signal from [1] with (2).

Proof. The required fourth-moment estimate uses $\kappa = 3/4$, because its zero-free input is $\beta_* \leq (1 + \kappa)/2 = 7/8$. Every detected bin has $\delta = 2a - 1 \leq 2\beta_* - 1 \leq 3/4$. Its plain prime capacity is thus the baseline capacity in the proof of [1, Part II, capacity comparison], with no positive $\Delta/4$ surcharge. The row exponent is bounded by $R_*(\delta, x)$ plus an arbitrarily small loss, with $0 \leq x \leq 1/2$ and $q = x\delta$. The same selected-prime argument applies since $\ell/h > 8/39 > 7/37$ and $5\ell - h > 1/48$. A fixed sufficiently fine even slot mesh meets all rounding conditions.

At row length $d = h$, the general high exponent relative to $C(B)$ is

$$E_B(h) = a - B + h \left(\frac{17}{50} - \frac{1}{6} \right) - a\ell_y - \frac{\ell}{2} + q\ell + h \left(R + \frac{\delta}{2} - \frac{17}{50} \right). \quad (5)$$

The endpoint certificate of [1, Part II, compensated endpoint certificate], evaluated at its original geometry, bounds this expression with $B = 7/8$ and $R = R_*$ by $-49/440640$ for the larger range $\delta \leq 5/6$. Holding a, δ, q, R fixed while changing (1) gives

$$E_B(h) - E_{7/8}(13/16) = \frac{1}{200000} + t \left(-\frac{1}{2} + \delta + q + \frac{3R}{2} \right). \quad (6)$$

The source bounds $R_* \leq 17/12$. Since $\delta \leq 3/4$ and $q \leq \delta/2$, the right side of (6) is at most

$$\frac{1}{200000} + \frac{11t}{4} = \frac{7}{80000}.$$

Thus the central endpoint retains the explicit margin

$$-E_B(h) \geq \frac{49}{440640} - \frac{7}{80000} > 0. \quad (7)$$

The frequency slope is positive, so this endpoint bounds all moderate rows with $d \leq h$. Extending to $h + \zeta$ costs at most 2ζ , and ζ can be chosen below the margin in (7).

The floor and intermediate-row margins in [1] are respectively $7/1200$ and $49/14400$ at the original geometry. The floor shift costs $t(121/40 + 3e + \text{mesh})$, including the changed row threshold. The parameter budget absorbs this cost. The small-row margin is $63/800$. Its change is at most $51t/100$, since $h(17/50 - 1/6) - l_y/2$ has derivative $51/100$ with respect to t . For large rows, the absolute contour can still be taken sufficiently far right after choosing ζ . Lemma 3 supplies the principal contour, whose margins $l_y/20$ and $h/600$ remain positive. Its local approximation supplies a positive power saving in each fixed slot. The prime normalizer costs only $Z^{o(1)}$.

All detector, mesh, real-contour, and normalizer losses can be chosen below these fixed positive margins before the target character is fixed. The late height choice in [1] then gives one $\sigma > 0$ for every target, as claimed. $\square$

Proof of Theorem 1. Suppose $\beta_* > B$. Lemma 3 supplies a principal factor holomorphic and bounded away from zero on $\text{Re } s > B$. Lemma 2 and (4) give the direct bound required by the continuation criterion of [1, Section 2]. Its parameter ω can be any sufficiently small positive number below $\beta_* - B$. Lemma 4 gives the other required bound with a target-independent saving. The criterion contradicts $\beta_* > B$. Hence $\beta_* \leq B$. The transfer in [1] from primitive Hecke characters to imprimitive Hecke and Dirichlet characters proves the stated assertions. $\square$

Verification scope

The rational inequalities in (4) and (7) can be checked independently by the accompanying `verify_bounds.py`. The accompanying Lean files verify the geometry, reflected-exponent calculation, high-side endpoint, normalized compensated low estimate, local Euler bounds, and principal and global Euler product extensions against the upstream OAI formalization. The finite source multiplier bounds and the principal source's w and z contour shifts, ordered double shift, uniform contour tails, and x contour transport also have Lean proofs at the new boundary. The selected high-row local factors, weighted prime-slot sum, physical-row dyad, Mellin integral, and changed small- and large-row tail bounds, central cube truncation error, canonical ray-cube choice, and normalized floor-row collection are verified there as well. The principal slot-ratio and normalized residue estimates are proved on the new boundary. The physical principal remainder at the changed lengths is bounded on a fixed contour just right of $7/8$, leaving a power saving of $1/3000$ relative to β_* in the proposed interval. The balanced

central-bin estimate, principal transport, detector assembly, and continuation contradiction are formalized in Lean conditional on the stated chosen fourth-moment input. That input remains open. The final Lean theorem also takes $\beta_* \leq 7/8$ as a hypothesis, since the upstream certified-band module was not rebuilt in this checkout. Compiling this note checks the TeX, not the remaining number-theoretic inputs. The conditional stronger boundary in Theorem 1 is not part of [1]’s stated theorem or its existing formalization.

References

- [1] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\operatorname{Re}(s) > 7/8$, 30 September 2026. Source consulted 7 October 2026. [Source TeX](#).